

Strategic Data Ordering: Enhancing Large Language Model Performance through Curriculum Learning

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Abstract

The rapid advancement of Large Language Models (LLMs) has improved text understanding and generation but poses challenges in computational resources. This study proposes a curriculum learning-inspired, data-centric training strategy that begins with simpler tasks and progresses to more complex ones, using criteria such as prompt length, attention scores, and loss values to structure the training data. Experiments with Mistral-7B (Jiang et al., 2023) and Gemma-7B (Team et al., 2024) models demonstrate that curriculum learning slightly improves performance compared to traditional random data shuffling. Notably, we observed that sorting data based on our proposed attention criteria generally led to better performance. This approach offers a sustainable method to enhance LLM performance without increasing model size or dataset volume, addressing scalability challenges in LLM training.

1 Introduction

The rapid advancement of Large Language Models (LLMs) has been a defining trend in artificial intelligence, reshaping the capabilities of machines in understanding and generating human-like text. This surge in progress can be traced back to the introduction of the Transformer architecture (Vaswani et al., 2017), which laid the groundwork for the development of models such as the Generative Pre-trained Transformer (GPT) family. These models have been pivotal in pushing the boundaries of what machines can achieve, largely due to their expansive parameter counts and sophisticated training methodologies. The Transformer’s introduction has not only catalyzed significant progress but also sparked a paradigm shift in how we approach machine learning tasks (Acheampong et al., 2021; Patwardhan et al., 2023; Ali et al., 2024).

OpenAI’s research underscores a key observation in the progression of these models: larger models tend to exhibit superior performance (Kaplan

et al., 2020). However, this approach of scaling up models introduces a set of challenges, notably the increased demand for computational resources and associated costs. As the size of models escalates, it prompts a crucial question about the sustainability of such growth trajectories and whether alternative strategies might offer more efficient pathways to improving model performance.

In response to these challenges, we propose a data-centric training strategy, inspired by the pedagogical concept of curriculum learning. Curriculum learning, a strategy that mimics human educational approaches by starting with simpler tasks and progressively moving to more complex ones. By structuring the training data and tasks in a manner that reflects the incremental learning stages humans go through, this method aims to facilitate a more natural and efficient learning process for the models. This approach deviates from traditional training practices, which often involve exposing models to a random assortment of data, suggesting a structured progression could yield better performance and efficiency.

The potential of curriculum learning in the realm of LLMs is profound. It not only aligns with cognitive and educational theories but also presents a practical framework to tackle the scalability challenges associated with model training. By focusing on the complexity of training data and adopting a gradual learning, curriculum learning could lead to the better performance without the necessity for constant increases in model size.

Therefore, we focus on data-centric strategy training method by curriculum learning, by examining the impact of performance of model, comparing random shuffled training dataset.

This study focuses on a data-centric strategy to enhance model performance through the adoption of curriculum learning. By examining the impact of structured training data, this study aims to demonstrate its effectiveness relative to the traditional

method of random shuffling. We seek to establish that curriculum learning is a crucial strategy for improving the development process of large language models, potentially achieving better performance without increasing the model parameters or the size of the dataset.

2 Related Work

Curriculum learning, introduced by Bengio et al. in 2009 (Bengio et al., 2009), is a learning strategy where models are trained on data from simple to complex, mirroring human learning processes. This approach enhances model performance by establishing a foundational understanding before introducing more complex data. It has been widely applied in natural language processing (NLP) and computer vision. (Guo et al., 2018; Antonios Platanios et al., 2019; Tay et al., 2019).

In NLP, curriculum learning has proven to be effective by organizing the training sequence to begin with simpler texts and progressively transition to more complex ones. The complexity of these texts is assessed based on various criteria, including text length, the rarity of vocabulary, and edit distance. This structured approach has successfully demonstrated the efficacy of curriculum learning in the training of Pretrained Language Models (PLMs) (Chang et al., 2021; Nagatsuka et al., 2023).

Recent advancements in large-scale language and multimodal models have highlighted the effective integration of curriculum learning with cutting-edge techniques, significantly boosting performance across various applications. Notably, projects such as WavLLM (Hu et al., 2024) and AutoWebGLM (Lai et al., 2024) showcase the potential of these integrations, tailoring the learning process to gradually introduce more complex tasks, thereby enhancing model robustness and application specificity. Alongside these innovations, further research has validated the effectiveness of curriculum learning in instruction tuning for language models. This approach not only systematically improves models’ abilities to handle progressively challenging tasks but also optimizes their learning trajectory for better performance in specialized tasks. By methodically structuring the training process, these initiatives have set a new standard in the development and refinement of AI models, marking a significant step forward in their evolution.

In the development of training data for curricu-

lum learning, accurately measuring data difficulty is crucial. Specifically, for LLMs (Large Language Models), determining the degree of data difficulty is challenging. Traditional metrics, such as text length or word rarity, are often employed to estimate the difficulty of training data (Chang et al., 2021; Nagatsuka et al., 2023). However, these metrics may not fully reflect the complexity of a dataset. It is essential to assess data difficulty from the model’s perspective, rather than relying solely on data-specific metrics. Our research proposes a new approach to calculate the degree of difficulty based on a model-centric perspective. By organizing the training dataset according to difficulty using our new metric, we aim to improve the model’s performance compared to random shuffling.

3 Methods for Quantitative Difficulty Measurement

In this section, we introduce a novel methodology for training models that begins with easier tasks and methodically progresses to more challenging ones. This approach requires organizing data by its level of complexity, for which we have established three principal criteria. By arranging the data in an order that goes from less to more difficult, this approach establishes a structured progression for learning.

Length

It is widely acknowledged in the study that longer sentences are more challenging to learn. To assess difficulty based on length, we address the measurement of length in tokenized prompt, a critical factor in understanding the complexities of language processing. The prompt length, denoted as $d_{\text{length}}(p)$, is defined as the total number of tokens produced by the tokenizer for a given prompt. Mathematically, it is represented as:

$$d_{\text{length}}(p) = N. \quad (1)$$

Attention Score

We introduce a novel method to determine the complexity of a prompt by analyzing model attention scores. In our approach, we suggest that how a model focuses on different parts of a prompt—reflected by attention scores—can show how difficult the prompt is. If a model concentrates its attention on just a few token parts, the prompt is likely easier since only those parts are needed to find the answer. However, if the attention is spread

evenly across the whole prompt, it’s harder, indicating that understanding everything is necessary to get the right answer. We measure this by looking at the variation in attention scores, using two specific methods.

This study focus exclusively on the attention scores for tokens following the section where the response is generated to analyze response generation. We calculate the variance of attention scores within each attention layer for these tokens and then average these variances across all layers. This process is mathematically represented as:

1. For each layer i , calculate the variance of attention scores, $\text{Var}(A_i)$, for tokens after the section where the response is generated.
2. Compute the average variance across all N layers:

$$d_{\text{Average Variance}} = \frac{1}{N} \sum_{i=1}^N \text{Var}(A_i)$$

Loss

A high loss signifies a substantial discrepancy between the predicted and actual outcomes, indicating that the model perceives the data as challenging. We establish a direct correlation between high loss values and increased difficulty. Therefore, prompts that generate higher loss are considered more challenging for the model to learn.

This method computes the cross-entropy loss associated with the answer segment of the prompt.

$$d_{\text{loss}} = - \sum_i \text{Prompt}_i \log(\text{Answer}_i) \quad (2)$$

4 Experiment Settings

Models

In order to validate our hypotheses, this study conducted a series of experiments utilizing language models with 7 billion parameters. Specifically, we utilized the Mistral-7B and Gemma-7B models to conduct our analyses (Jiang et al., 2023; Team et al., 2024).

In addition, because of limited resources, we utilized Quantize Low Rank Approximation (QLoRA) (Hu et al., 2021) with Parameter-Efficient Fine-Tuning (PEFT) library to fine-tune the models. Different from full fine-tuning, LoRA freezes the pre-trained model weights and injects learnable

rank decomposition matrices into each layer of the Transformer architecture. Thus, this method can reduce the number of trainable parameters, the cost of training and time for downstream tasks. The details of configurations are described on Table 1.

Instruction tuning Datasets

This study focus on specific datasets that are considered the state-of-art results for instruction tuning. Due to the memory limitations of our hardware setup, this study is limited to using single-turn datasets.

- **Orca-math** (Mitra et al., 2024) This dataset contains answers generated for elementary school math word problems using Azure GPT-4 Turbo.
- **Alpaca** (Taori et al., 2023) This dataset was created using the Self-Instruct (Wang et al., 2022) method and the OpenAI Text-Davinci-003 model.
- **SlimOrca-Dedup** (Lian et al., 2023b) This is a subset of the OpenOrca (Mukherjee et al., 2023) data, where, within the FLAN (Longpre et al., 2023) dataset, incorrect answers were identified and removed using OpenAI GPT-4 based on human annotations, and duplicates were eliminated in the SlimOrca (Lian et al., 2023a) dataset.

5 Evaluation

Language Model Evaluation Harness

In order to evaluate instruction-tuned model, we used Language Model Evaluation Harness (Gao et al., 2023) a framework for few-shot evaluation of language models. This method conducted ARC with 25 shot (Clark et al., 2018), HellaSwag with 10 shots (Zellers et al., 2019), MMLU with 5 shot (Hendrycks et al., 2021), TrustfulQA with a 0 shot (Lin et al., 2021), Winogrande with 5 shot (ai2, 2019) and GSM8K with 5 shot (Cobbe et al., 2021) datasets. The combination of these datasets extensively evaluates the ability of Large Language Models to respond to realistic questions and reasoning. By evaluating these datasets, we verify whether our proposed curriculum learning method impacts the model outcomes.